

The background features three vertical stripes on the left: a wide pink stripe, a medium blue stripe, and a narrow light beige stripe. The right side of the slide is white, decorated with a grid of small pink dots in the top right and bottom right corners.

UNDERGRADUATE PROJECT

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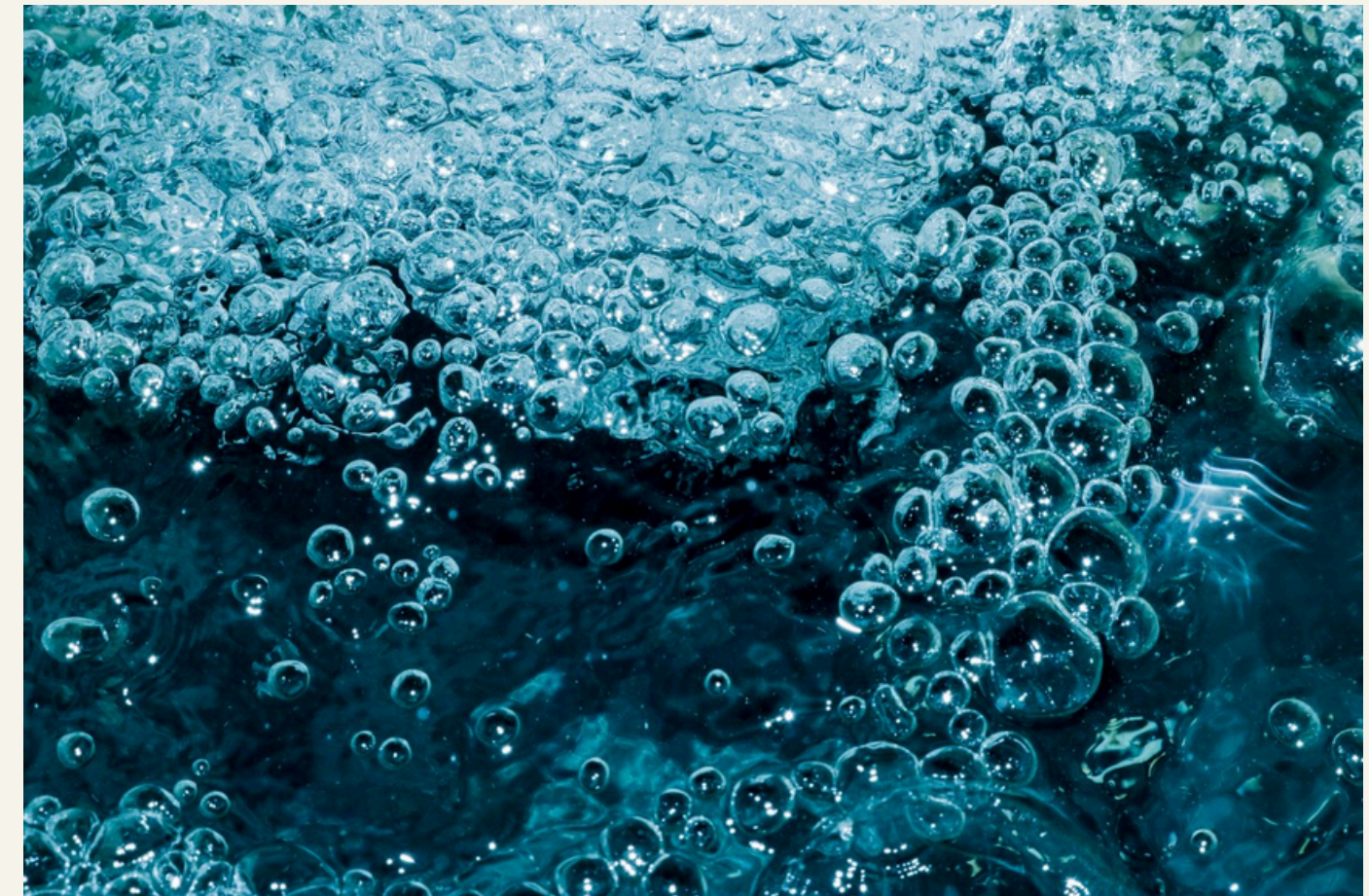
Under the guidance of : Prof DIPIN S PILLAI

OBJECTIVE

- In this project, we aim to predict the droplet size in a multiphase fluid flow system focusing on the behavior of an oil-water flow under a wide range of fluid flow conditions, for which we need to predict certain flow parameters
- To generate a standardized large-scale data set for oil-water flow system using COMSOL simulations.
- To develop a 3D CNN model capable of predicting the flow parameters of the system from the data set

INTRODUCTION

- **Multi Phase fluid flow is defined as simultaneous flow of two or more distinct phases, such as gas-liquid, liquid-liquid, or solid-liquid, within a system.**
- **The complexity of multiphase flow arises from the interactions between the different phases and the influence of various physical parameters, such as flow velocity, viscosity, density, and surface tension.**

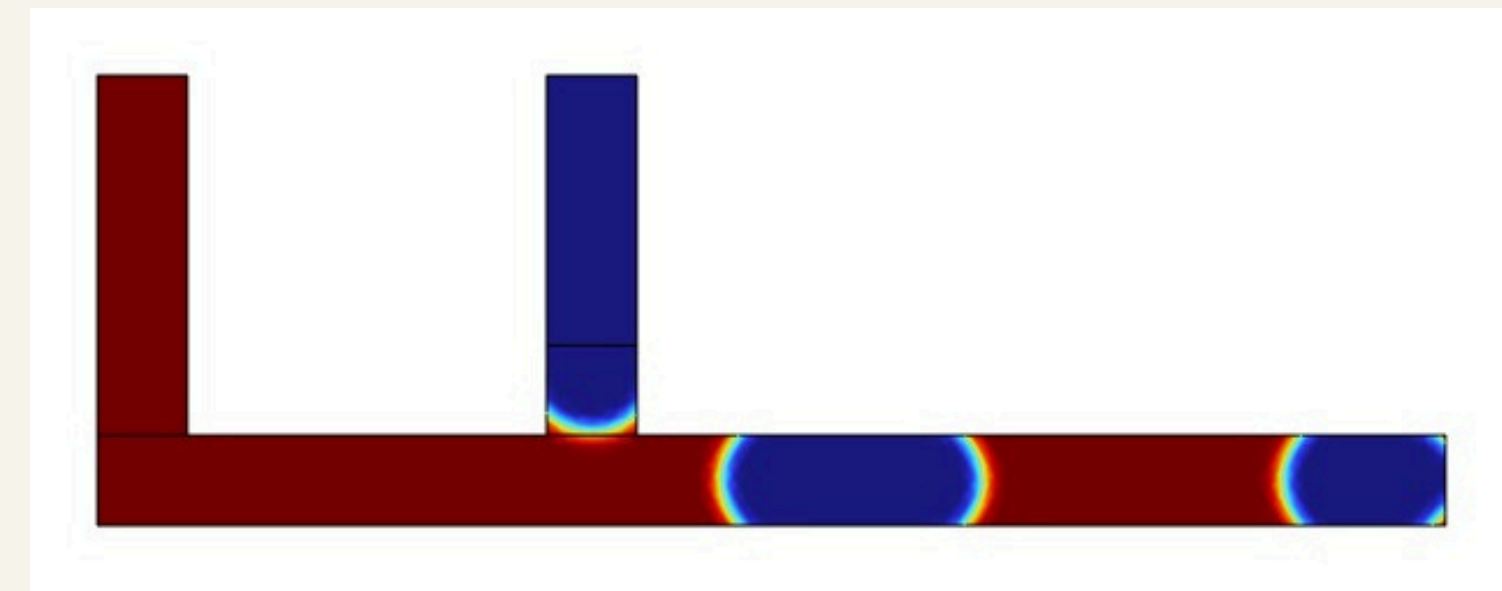
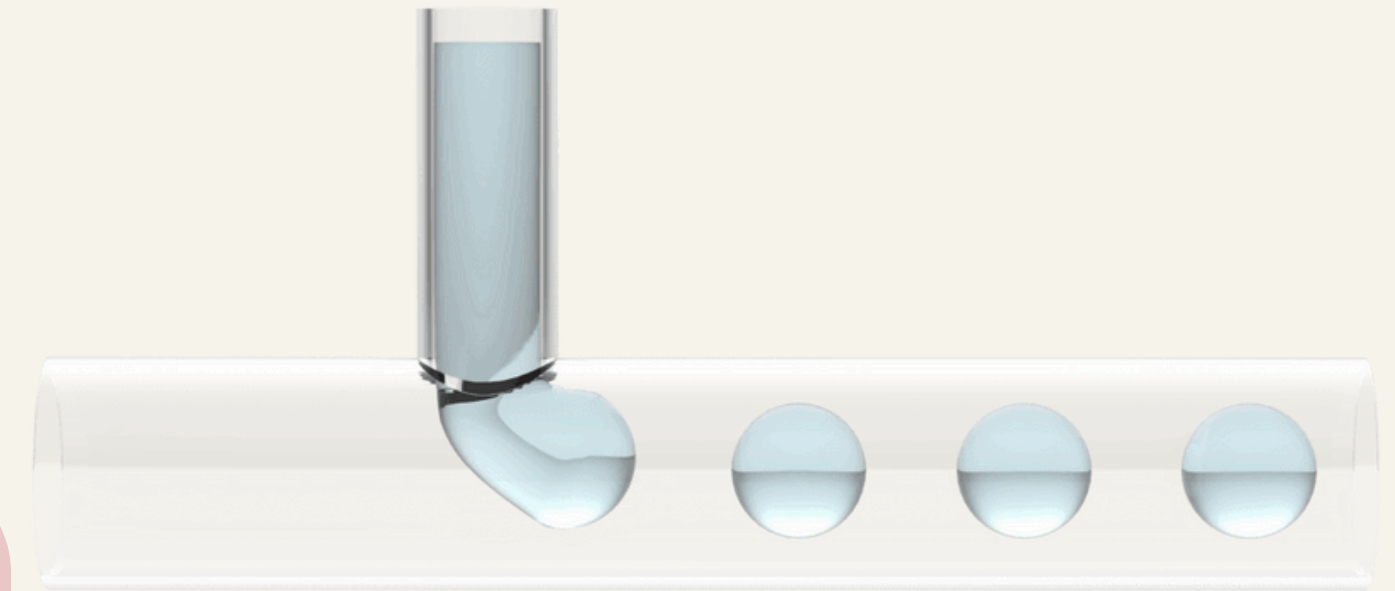


FLUID-FUID FLOW IN A T-JUNCTION

In a T-junction, two different fluid streams meet at a 90-degree angle, resulting in a complex interaction between the phases..

We focus on Dispersed Flow in Multiphase Systems where one phase (typically the dispersed phase) breaks into small droplets or bubbles within the other phase (the continuous phase)

The flow behavior is heavily influenced by the inlet flow rates, the physical properties of the fluids (such as viscosity, density, and surface tension), and the geometry of the junction.



ESTIMATED QUANTITIES

Flow rate ratio

Flow rate ratio is the ratio of flow rates of dispersed phase upon continuous phase in the two phase medium

$$\frac{Q_1}{Q_2} = \frac{A_1 V_1}{A_2 V_2} = \frac{V_1}{V_2}$$

Weber Number

The Weber number is the ratio of inertial forces to surface tension forces plays a critical role in determining the behavior of the dispersed phase

$$We = \frac{\rho v^2 L}{\sigma}$$

Capillary Number

The capillary number is the ratio of viscous forces to surface tension forces and is a key factor in predicting droplet behavior.

$$Ca = \frac{\mu v}{\sigma}$$

COMSOL SPECIFICATIONS:

MATERIAL PROPERTIES:

Density of water: 1000 Kg/m³

Density of oil: 1610 Kg/m³

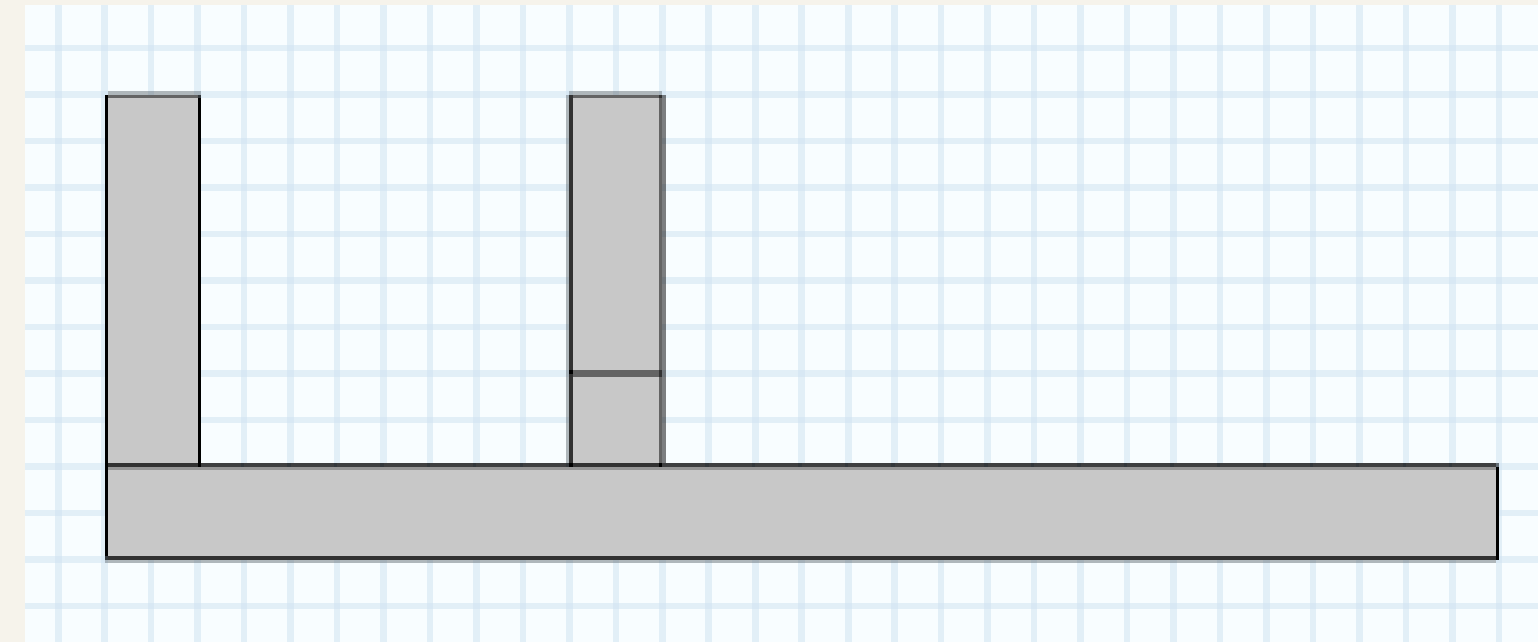
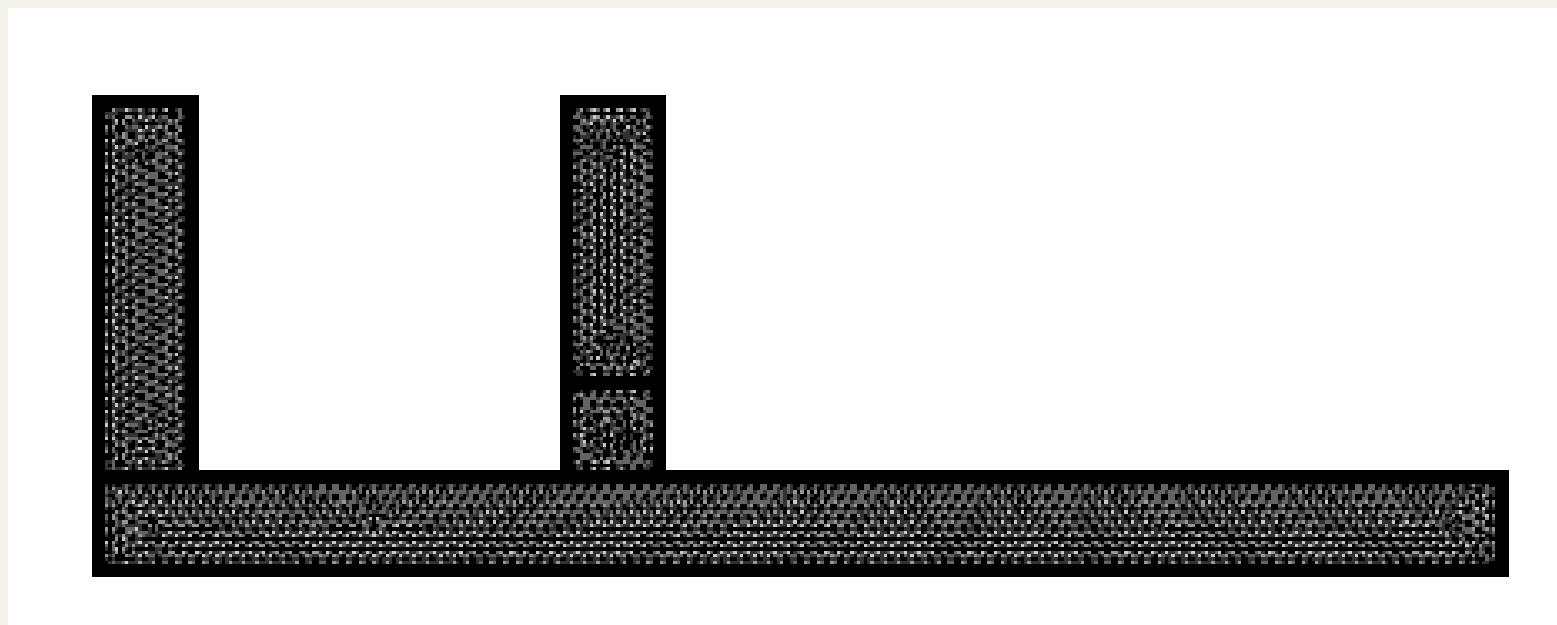
Viscosity of water: 0.001 Kg/ms

Viscosity of oil: 1.24×10^{-3} Kg/ms

Continuous Phase: Oil

Dispersed phase: Water

Contact Angle: $3\pi/4$



GEOMETRY & IMAGE SPECIFICATIONS:

Width of pipe(horizontal): 3 mm

Height of pipe(horizontal): 0.2 mm

Width of pipe(vertical): 0.2 mm

Height of pipe(vertical): 0.8 mm

Length of channel: 3 mm

Aspect Ratio: 640 x 480

Number of frames: 21

Multi phase Model used: Incompressible Flow

Physics Interface: Two Phase Flow, Level Set 1

Mesh: Fine mesh

USE CASES

We explored two scenarios by varying the flow parameters which are flow rate ratio ,weber number and capillary number

CASE-I

Keeping the capillary number constant , we varied the flow rate ratio and generated 300 data points where each data point is a set of 21 images, thereby training the model to predict flowrate ratio

CASE-II

All three parameters flow rate ratio, capillary number and weber number were varied to generate a data set of 1000 points where each data point is a set of 21 images.

METHODOLOGY

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Data Preparation

Data Pre-Processing

Model Creation

Model Training

Model Evaluation

The background features three vertical bars on the left: a wide pink one, a medium blue one, and a narrow beige one. The right side of the image has a light beige background with two rectangular areas of small, light pink dots in the top right and bottom right corners.

CASE 1

DATA PREPARATION

We leveraged COMSOL LiveLink for MATLAB to automate generation of simulations from COMSOL for different values of velocities and surface tension

```
% Setting velocity ranges
velocityRange1 = linspace(0.005, 0.05, 10);
velocityRange2 = linspace(0.0005, 0.005, 10);
```

```
% Setting current values of v1, v2, and surface tension
v1 = velocityRange1(i);
v2 = velocityRange2(j);
surfaceTension = 0.005*v2/0.002;
```

```
% Setting the inlet velocities and surface tension in the model and
% running the simulation for current values
model.physics('spf').feature('inl1').set('U0in', sprintf('%g[m/s]', v1));
model.physics('spf').feature('inl2').set('U0in', sprintf('%g[m/s]', v2));
model.multiphysics('tpf1').set('sigma', sprintf('%g[m/s]', surfaceTension));
model.study('std1').run();
```

```
% Calculate flow rate ratio, Weber number, and Capillary number and
% storing them in CSV file
flowRateRatio = v1 / v2;
weberNumber = (rho * (v1-v2)^2 * L) / surfaceTension;
capillaryNumber = (mu * v2) / surfaceTension;
fprintf(fileID, '%g,%g,%g,%g,%g,%g\n', v1, v2, surfaceTension, flowRateRatio, weberNumber, capillaryNumber);
```

```
% Exporting the simulation results as set of images
animation = model.result.export('anim1');
imageFileName = fullfile(targetFolder, sprintf('%04d_v1_%g_v2_%g_tension_%g.png', loopno, v1, v2, surfaceTension));
animation.set('imagefilename', imageFileName);
animation.run;
```

DATA PREPROCESSING

We imported our data in a Jupyter Source Notebook and used python to treat skewness in the labels and normalize the data. Then we split the data into train and test data with test size of 10%

1) Skewness Treatment of labels

```
df_transformed= np.log(df + 1)
```

2) Normalization of Labels

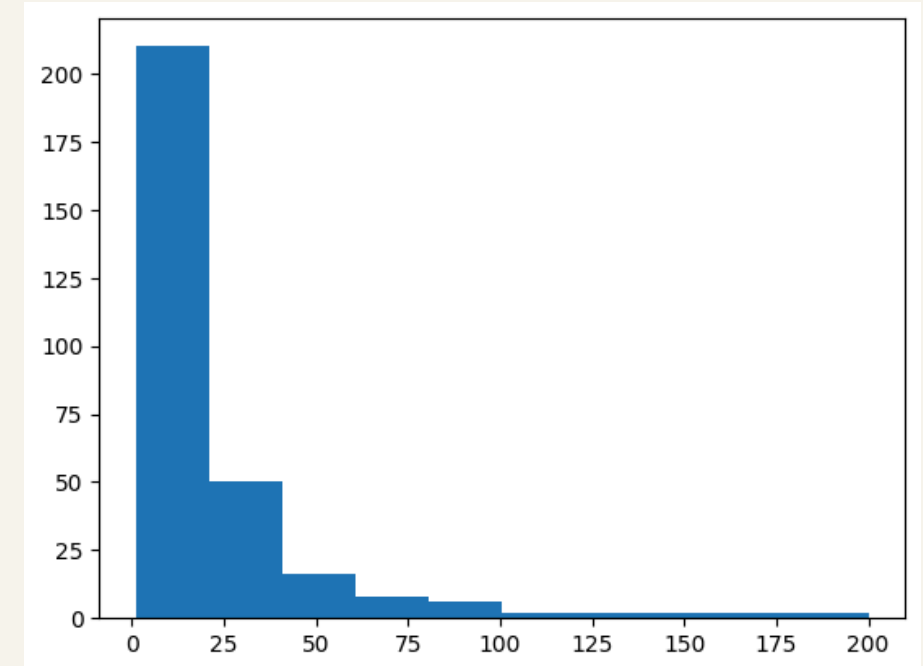
```
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
normalized_df = scaler.fit_transform(df_transformed.values.reshape(-1, 1))
```

3) Normalization of image data

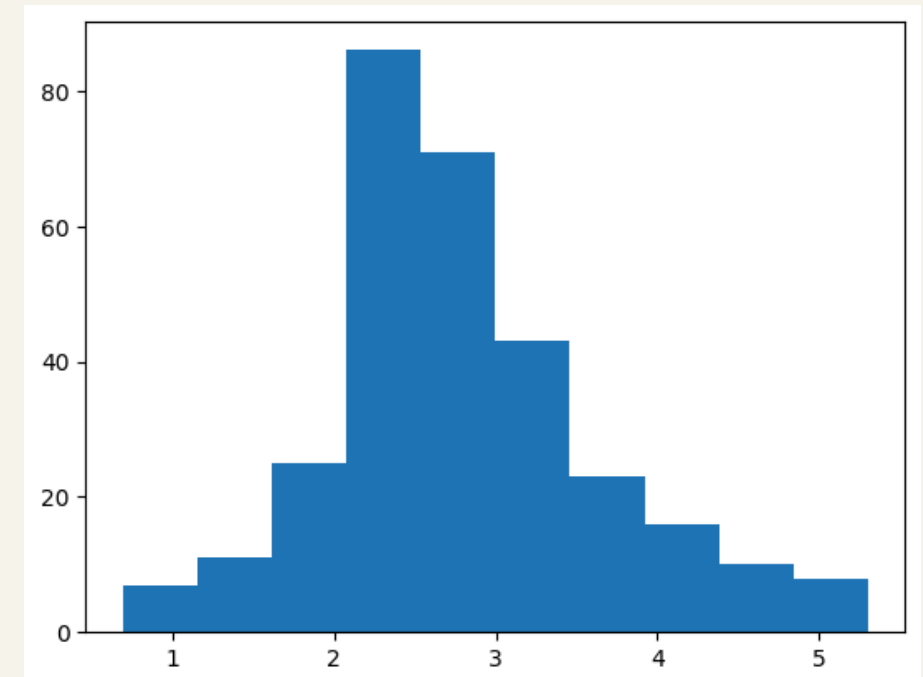
```
X = X.astype('float32') / 255.0
```

Splitting of train and test data

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.1, random_state=42)
```



Before preproession of labels



After preprocessing of labels

MODEL CREATION & TRAINING

We created a 3D CNN model and train it on train data using Adam optimizer with a learning rate of 0.001

```
# Initialising the Sequential model
model = Sequential()

# 3D Convolutional Layer 1
model.add(Conv3D(filters=16, kernel_size=(3, 3, 3), activation='relu',
| input_shape=(num_frames, img_height, img_width, num_channels)))
model.add(MaxPooling3D(pool_size=(1, 2, 2)))

# 3D Convolutional Layer 2
model.add(Conv3D(filters=32, kernel_size=(3, 3, 3), activation='relu'))
model.add(MaxPooling3D(pool_size=(1, 2, 2)))

# Flatten the output
model.add(Flatten())

# Dense Layers
model.add(Dense(64, activation='relu'))
model.add(Dense(32, activation='relu'))

# Output layer
model.add(Dense(1, activation='linear'))

model.compile(optimizer=Adam(learning_rate=0.001), loss='mse', metrics=['mae'])
model.summary()
```

```
history = model.fit(X_train, y_train, epochs=30, batch_size=2)
```

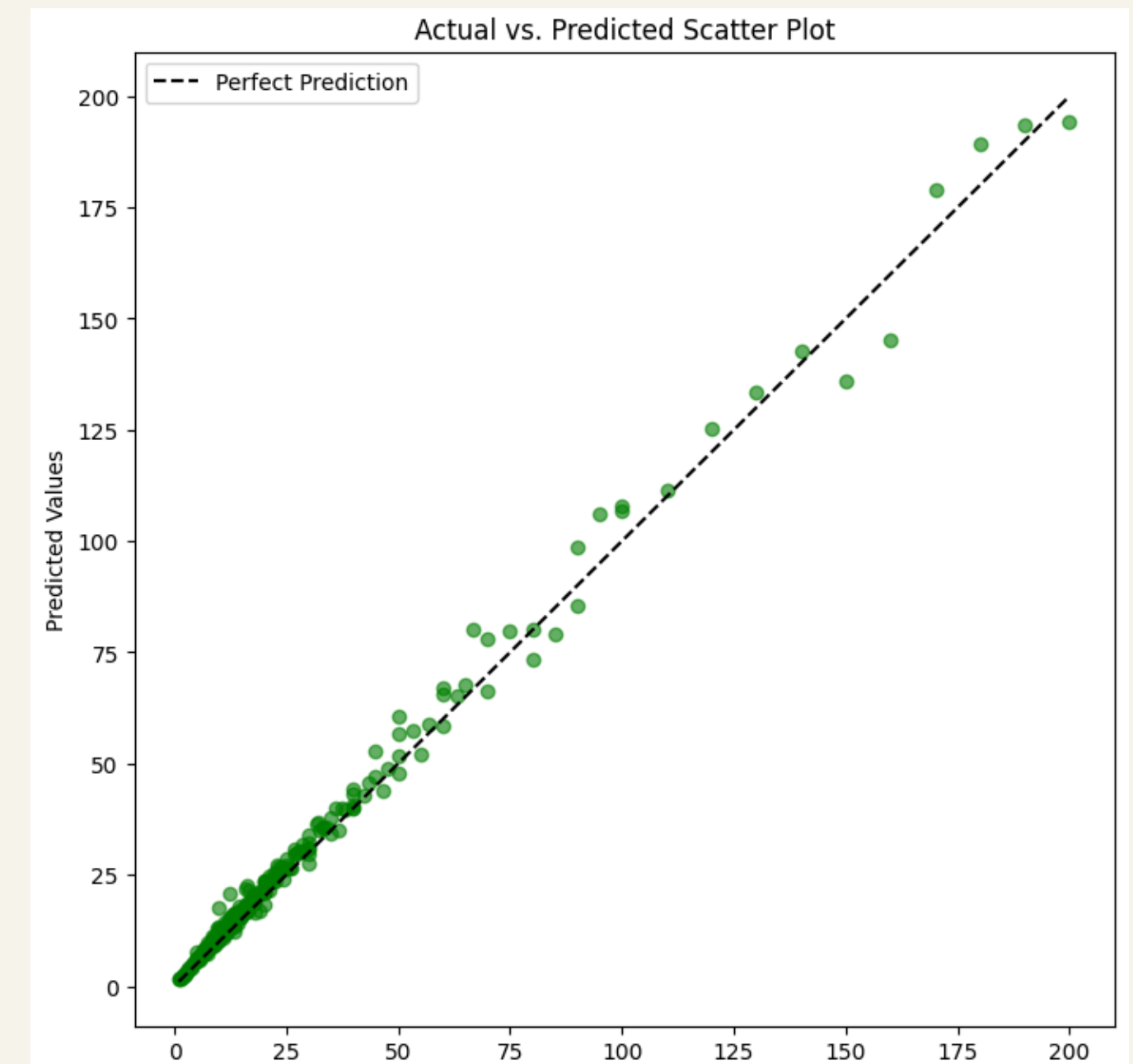
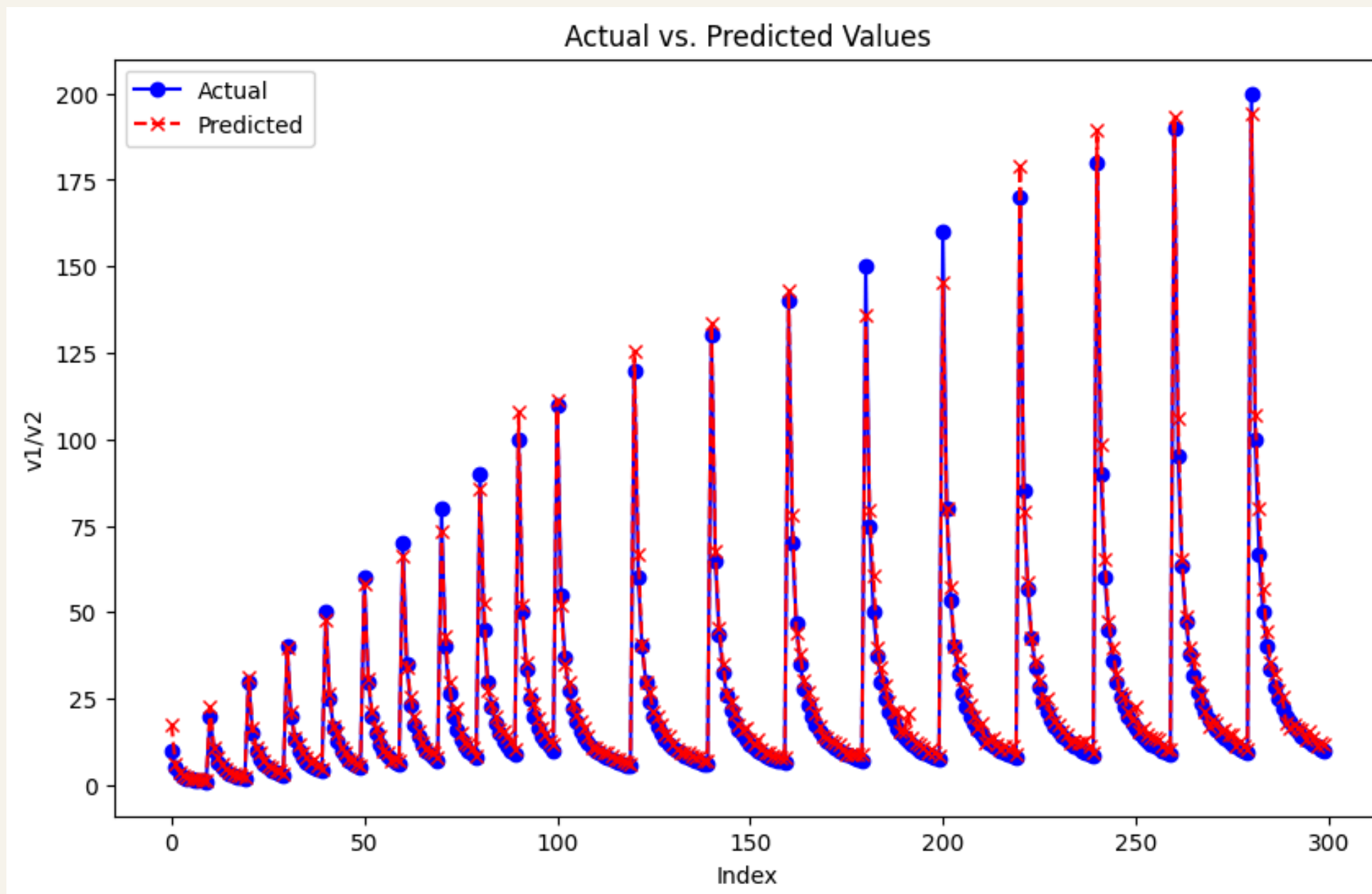
Input shape	(21,120,160,3)
No of Convolutional Layers	2
No of Dense Layers	3
Activation Function on Hidden Layers	ReLu
Activation Function on Output Layer	Linear
Optimiser	Adam
Learning Rate	0.001
No of Epochs	30

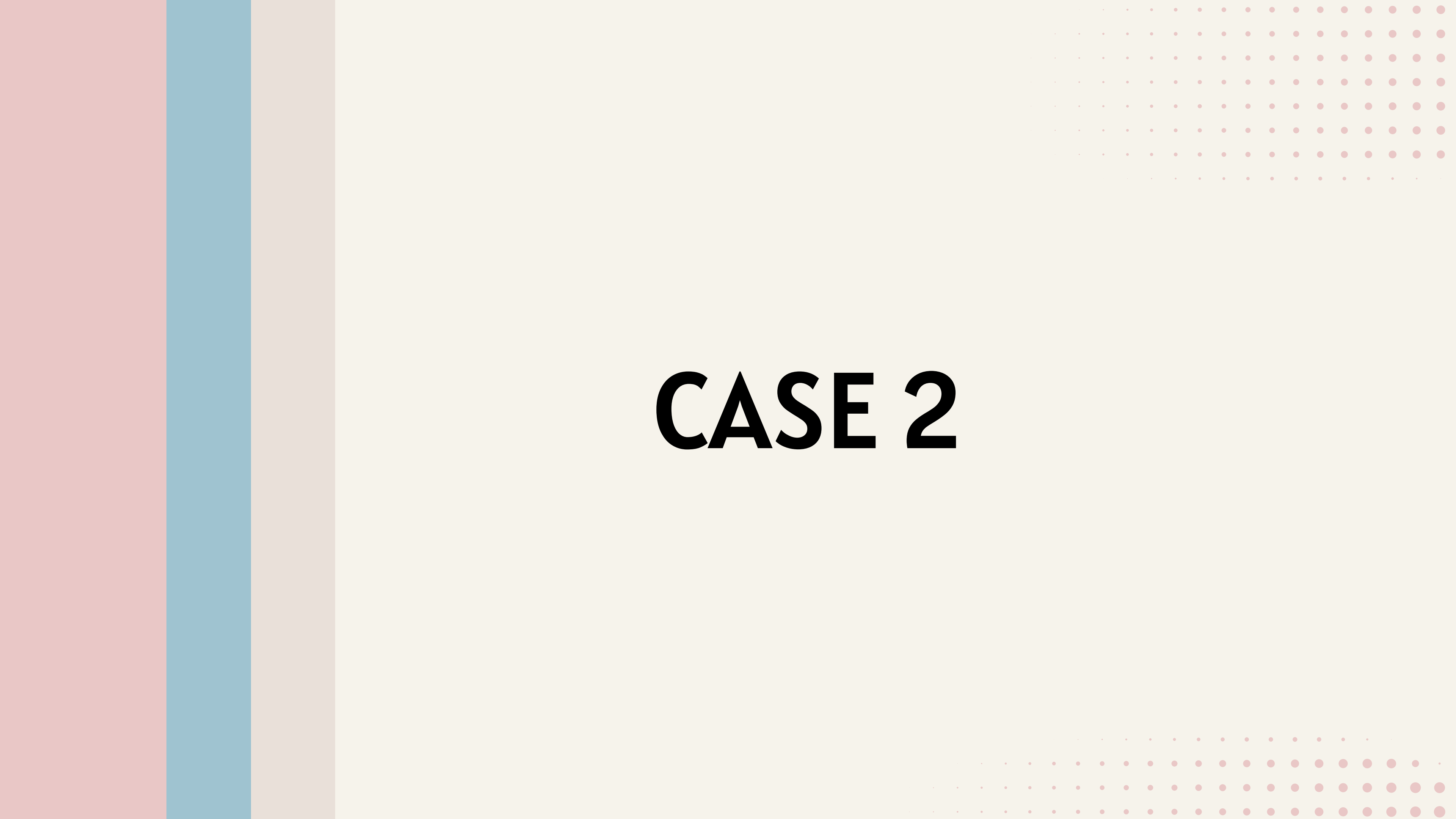
RESULTS

We calculated the R2 score of our model on train and test data respectively. Then we predicted the values of our data points using model and compared the predicted and actual values.

r2 score on train data: R^2 Score: 0.9794409814819383

r2 score on test data: R^2 Score: 0.9684441796366



The background features three vertical bars on the left: a wide pink bar, a medium blue bar, and a narrow beige bar. On the right side, there are two rectangular areas filled with a grid of small dots. The top area has a light beige background with pink dots, and the bottom area has a light pink background with pink dots.

CASE 2

DATA PREPARATION

We leveraged COMSOL LiveLink for MATLAB to automate generation of simulations from COMSOL for different values of velocities and surface tension

```
% Set velocity and surface tension ranges
velocityRange1 = linspace(0.005, 0.05, 10);
velocityRange2 = linspace(0.0005, 0.005, 10);
surfaceTensionRange = linspace(0.001, 0.01, 10);
```

```
% Set current values of v1, v2, and surface tension
v1 = velocityRange1(i);
v2 = velocityRange2(j);
surfaceTension = surfaceTensionRange(k);
```

```
% Set the inlet velocities in the model
% Run the simulation for the current values
model.physics('spf').feature('inl1').set('U0in', sprintf('%g[m/s]', v1));
model.physics('spf').feature('inl2').set('U0in', sprintf('%g[m/s]', v2));
model.multiphysics('tpf1').set('sigma', sprintf('%g[m/s]', surfaceTension));
model.study('std1').run();
```

```
% Calculate flow rate ratio, Weber number, and Capillary number
% Store the calculated values in the CSV file
flowRateRatio = v1 / v2;
weberNumber = (rho * (v1-v2)^2 * L) / surfaceTension;
capillaryNumber = (mu * v2) / surfaceTension;
fprintf(fileID, '%g,%g,%g,%g,%g,%g\n', v1, v2, surfaceTension, flowRateRatio, weberNumber, capillaryNumber);
```

```
% Export the simulation results as images with zero-padded filenames
animation = model.result.export('anim1');
imageFileName = fullfile(targetFolder, sprintf('%04d_v1_%03d_v2_%03d_tension_%03d.png', loopno, i, j, k));
animation.set('imagefilename', imageFileName);
animation.run;
```

We imported our data in a Jupyter Source Notebook and used python to treat skewness in the labels and normalize the data. Then we split the data into train and test data with test size of 15%

1) Skewness treatment of labels

```
for column in df.columns:  
    df_transformed[column]= np.log(df[column] + 1)
```

2) Normalization of labels

```
from sklearn.preprocessing import MinMaxScaler  
scalers = []  
for column in df_transformed.columns:  
    scaler = MinMaxScaler()  
    df_transformed[column] = scaler.fit_transform(df_transformed[[column]])  
    scalers.append(scaler)
```

3) Normalization of image data

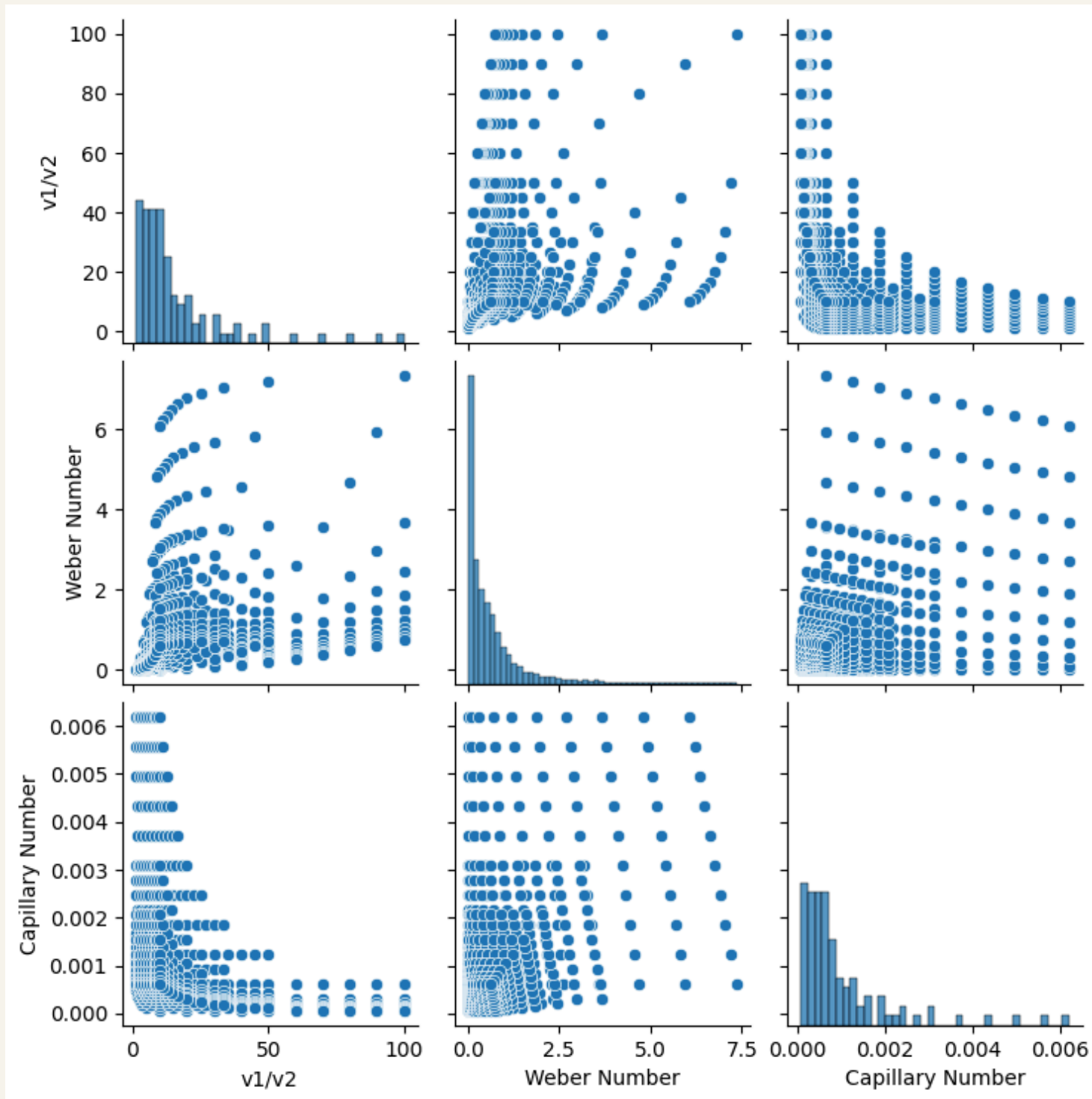
```
X = X.astype('float32') / 255.0
```

Splitting of train and test data

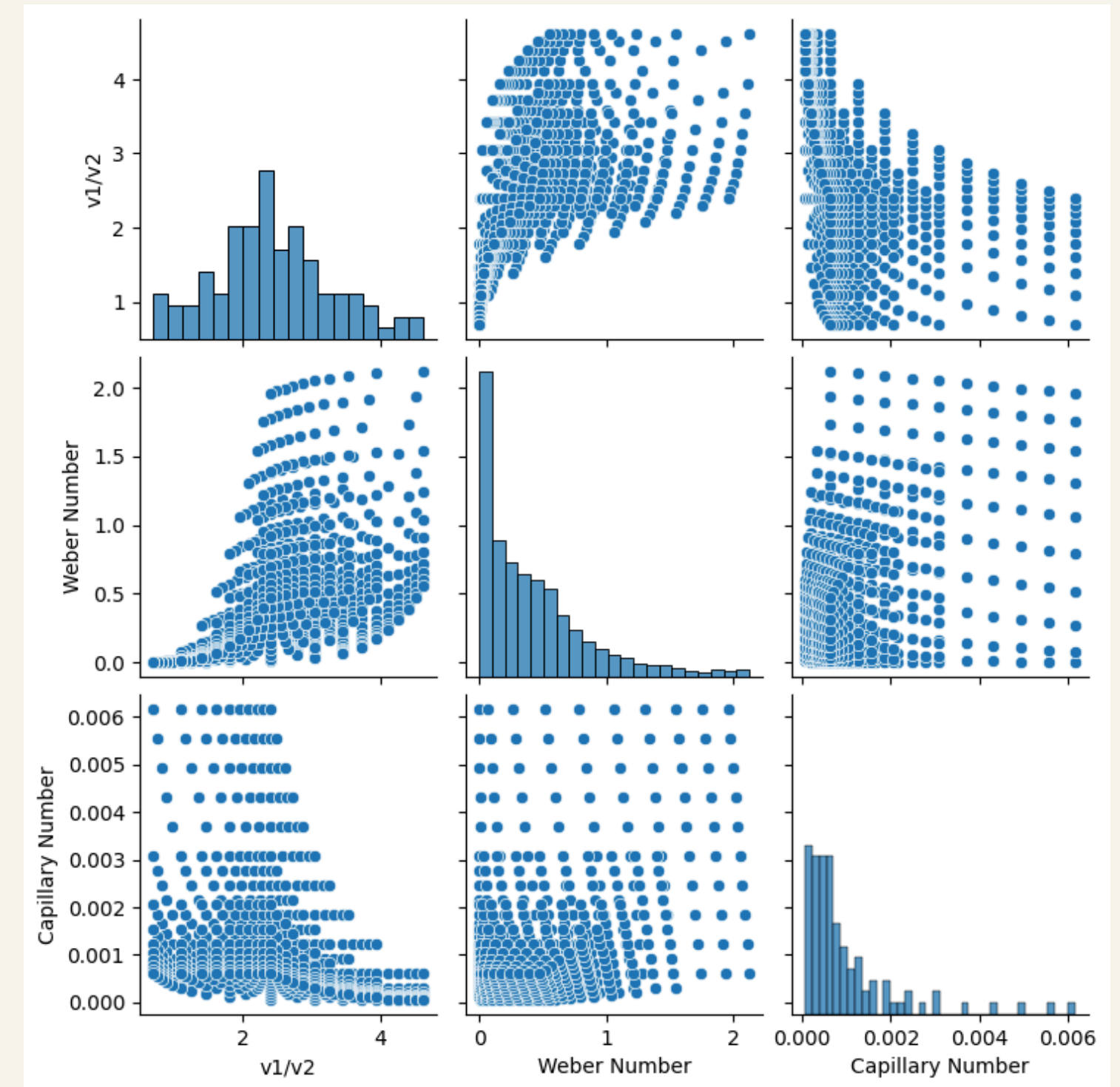
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.15, random_state=42)
```

DATA PREPROCESSING

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Before preprocessing of labels



After preprocessing of labels

MODEL CREATION & TRAINING

We created a 3D CNN model and train it on train data using Adam optimizer with a learning rate of 0.001

```
# Build the 3D CNN model
model = Sequential()

# 3D Convolutional Layer 1
model.add(Conv3D(filters=16, kernel_size=(3, 3, 3), activation='relu',
| input_shape=(num_frames, img_height, img_width, num_channels)))
model.add(MaxPooling3D(pool_size=(1, 2, 2)))

# 3D Convolutional Layer 2
model.add(Conv3D(filters=32, kernel_size=(3, 3, 3), activation='relu'))
model.add(MaxPooling3D(pool_size=(1, 2, 2)))

# Flatten the output
model.add(Flatten())

# Dense Layers
model.add(Dense(64, activation='relu'))
model.add(Dense(32, activation='relu'))

# Output layer for three independent labels
model.add(Dense(3))

model.compile(optimizer=Adam(learning_rate=0.001), loss='mse', metrics=['mae'])
model.summary()
```

```
history = model.fit(X_train, y_train, epochs=30, batch_size=2)
```

Input shape	(21,60,80,3)
No of Convolutional Layers	2
No of Dense Layers	3
Activation Function on Hidden Layers	ReLu
Activation Function on Output Layer	Linear
Optimiser	Adam
Learning Rate	0.001
No of Epochs	30

RESULTS

We calculated the R² score of our model on train and test data respectively. Then we predicted the values of our data points using model and compared the predicted and actual values.

Output 1	Flow Rate Ratio
Output 2	Weber No
Output 3	Capillary No

r² score on train data:

R² score for output 1: 0.9864085043492941

R² score for output 2: 0.9754189262576495

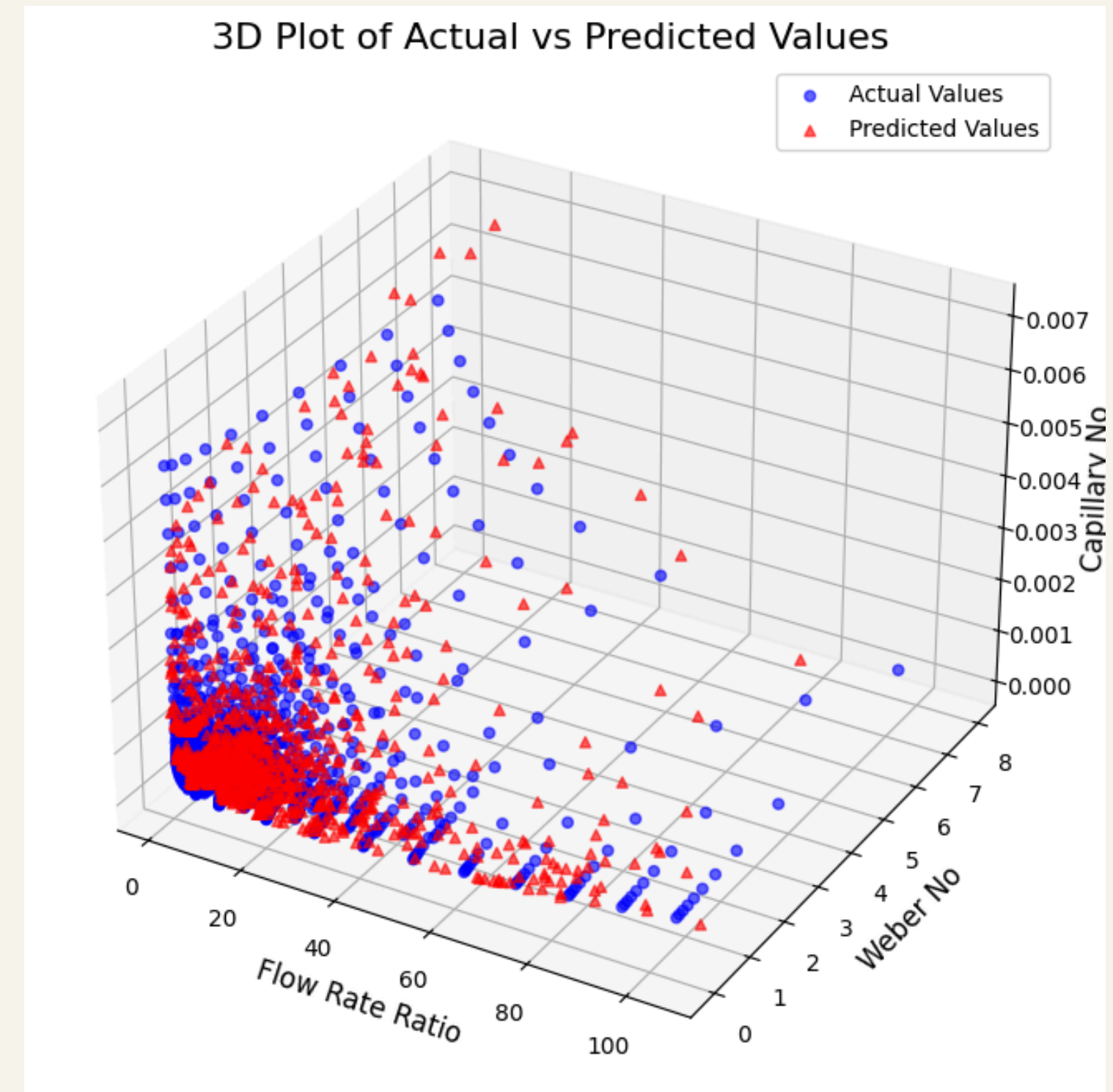
R² score for output 3: 0.9468829928195236

r² score on test data:

R² score for output 1: 0.9693266070697194

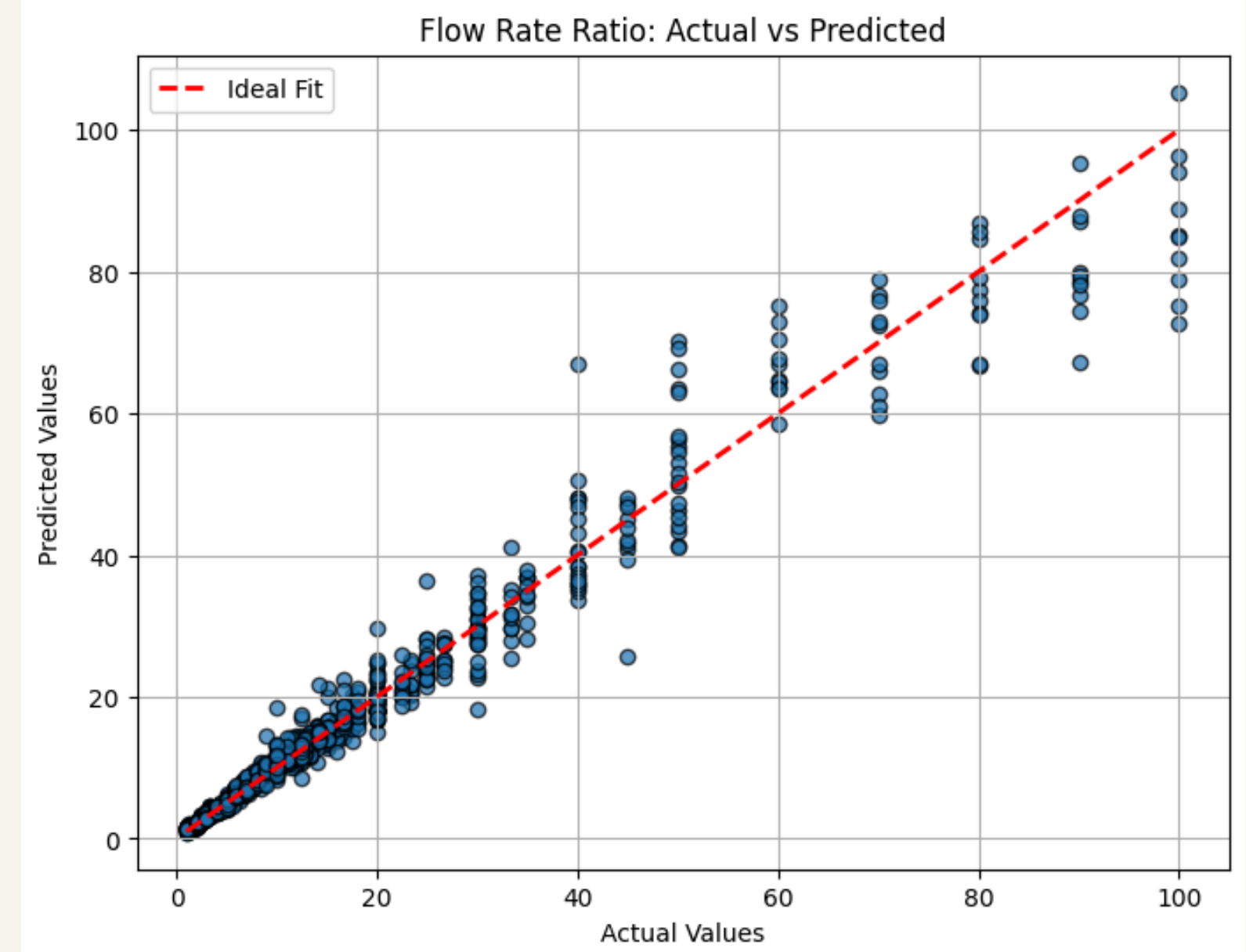
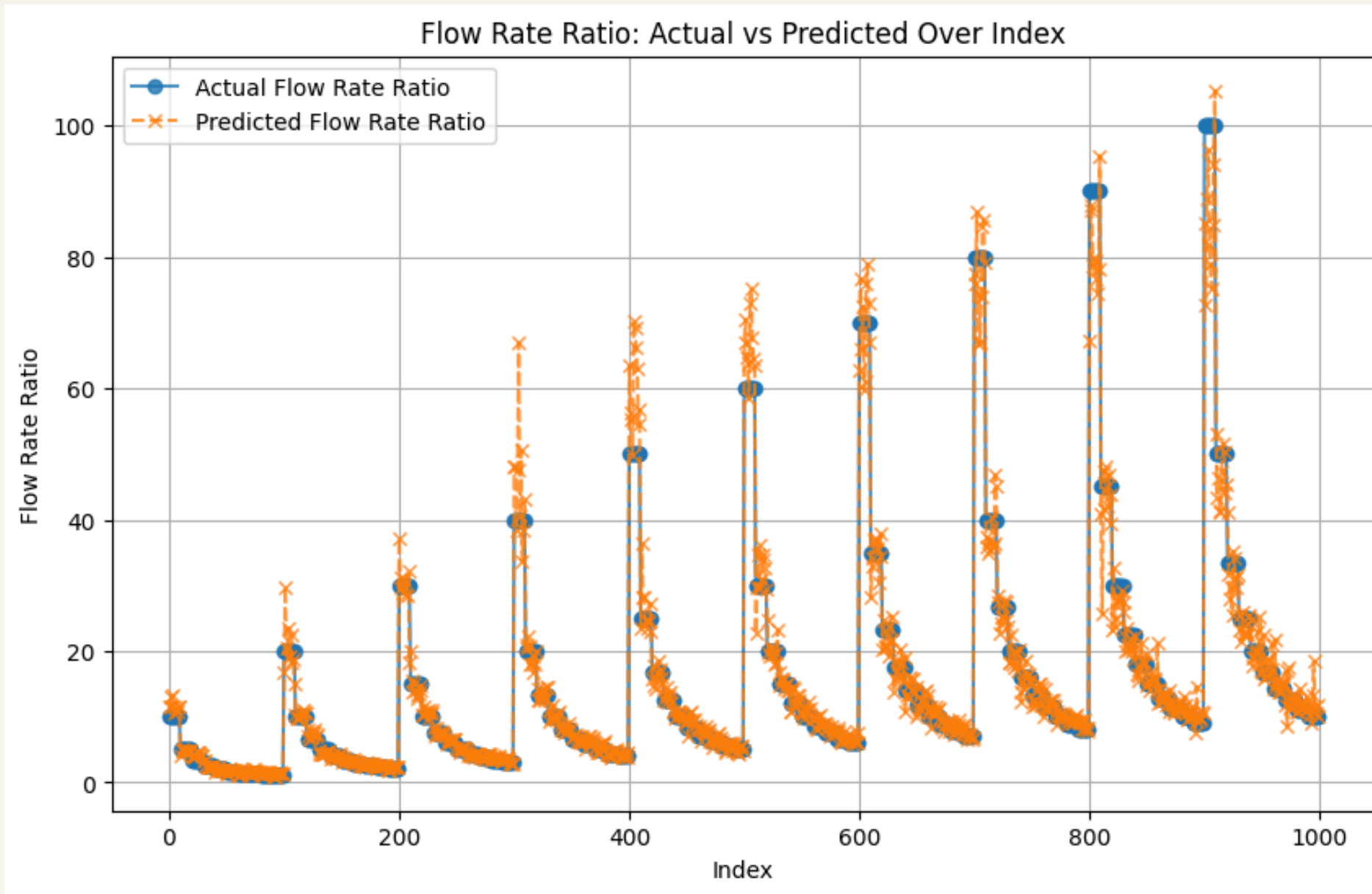
R² score for output 2: 0.92075549843047

R² score for output 3: 0.9357772592352885



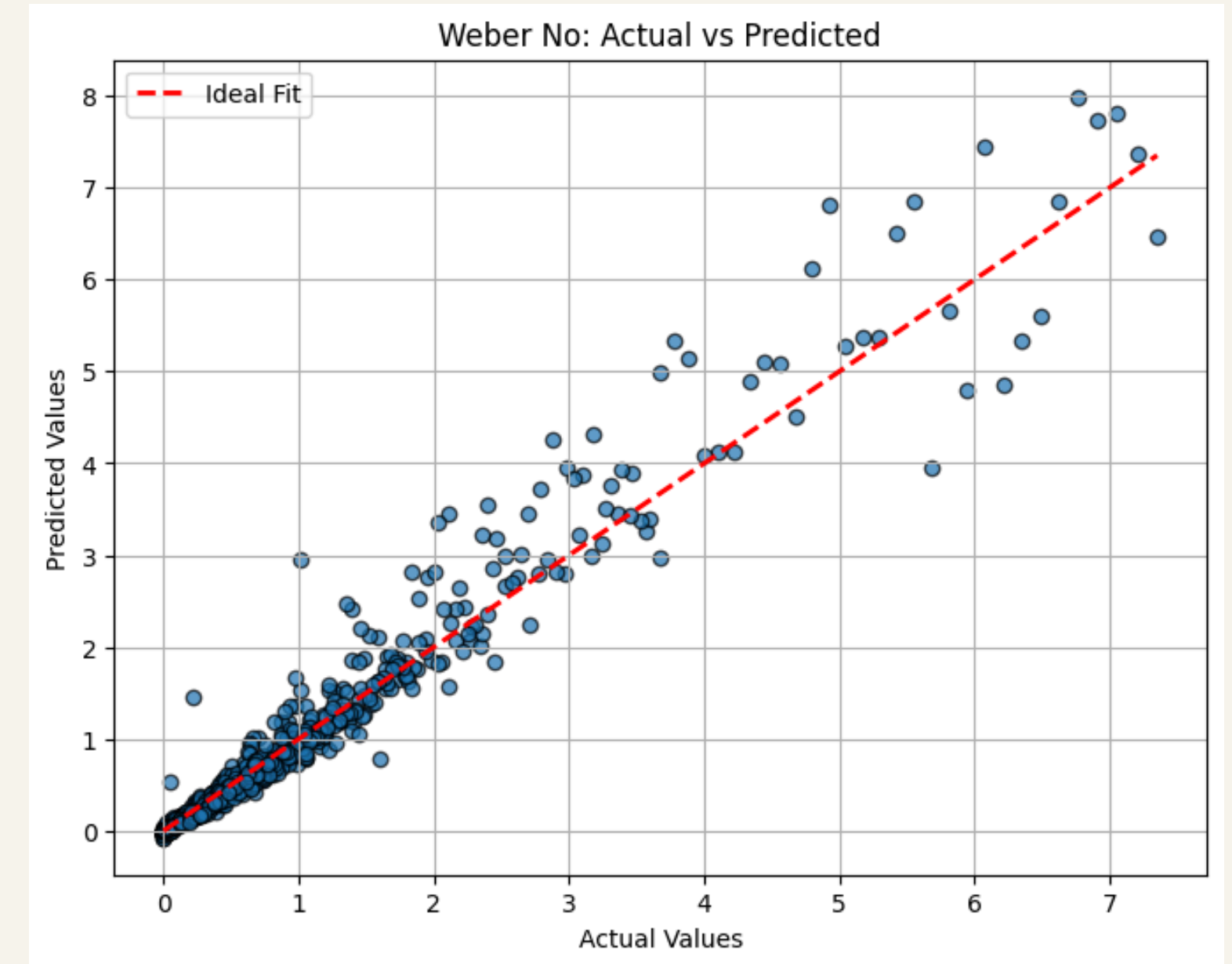
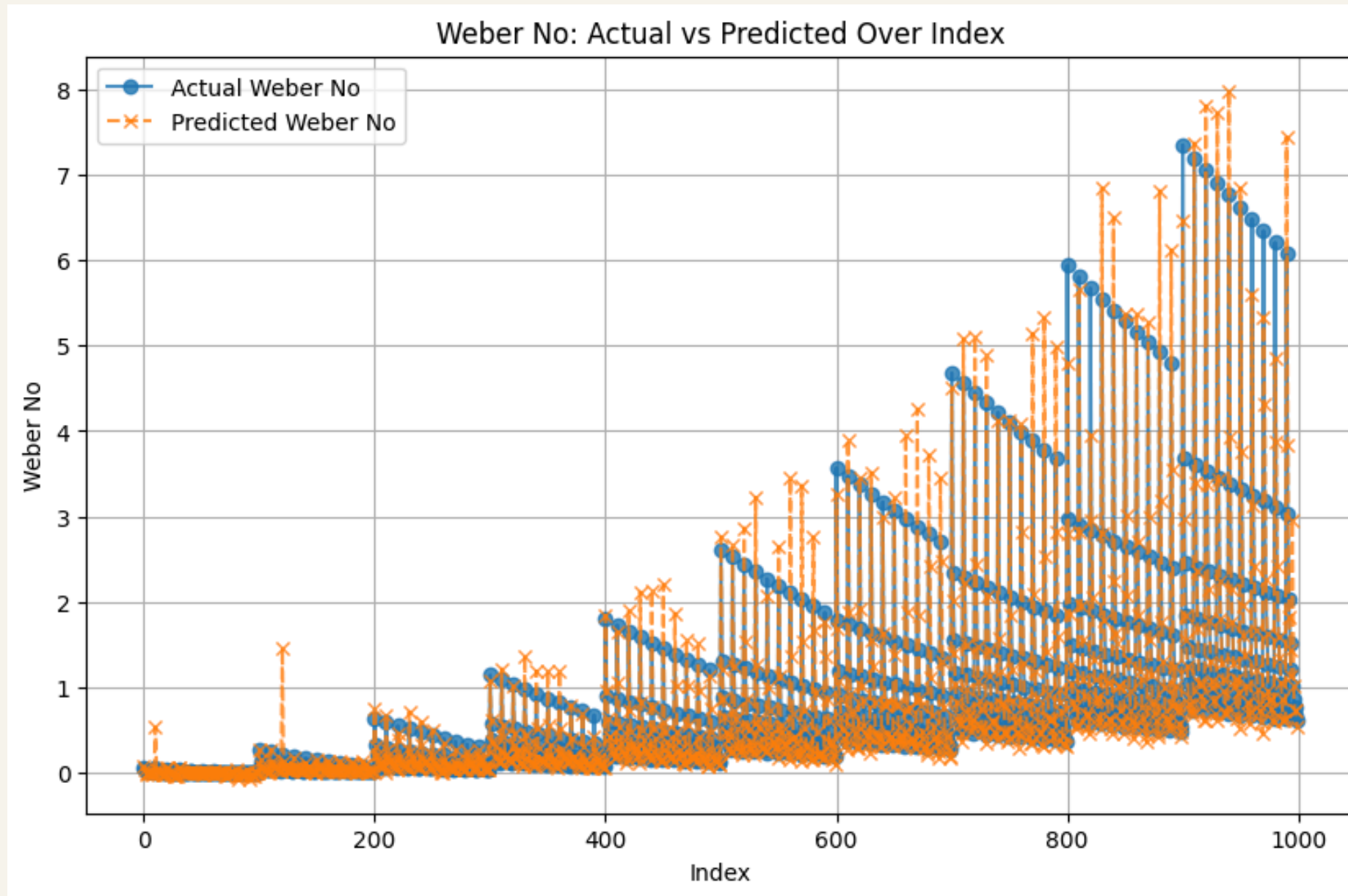
RESULTS - FLOW RATE RATIO

17



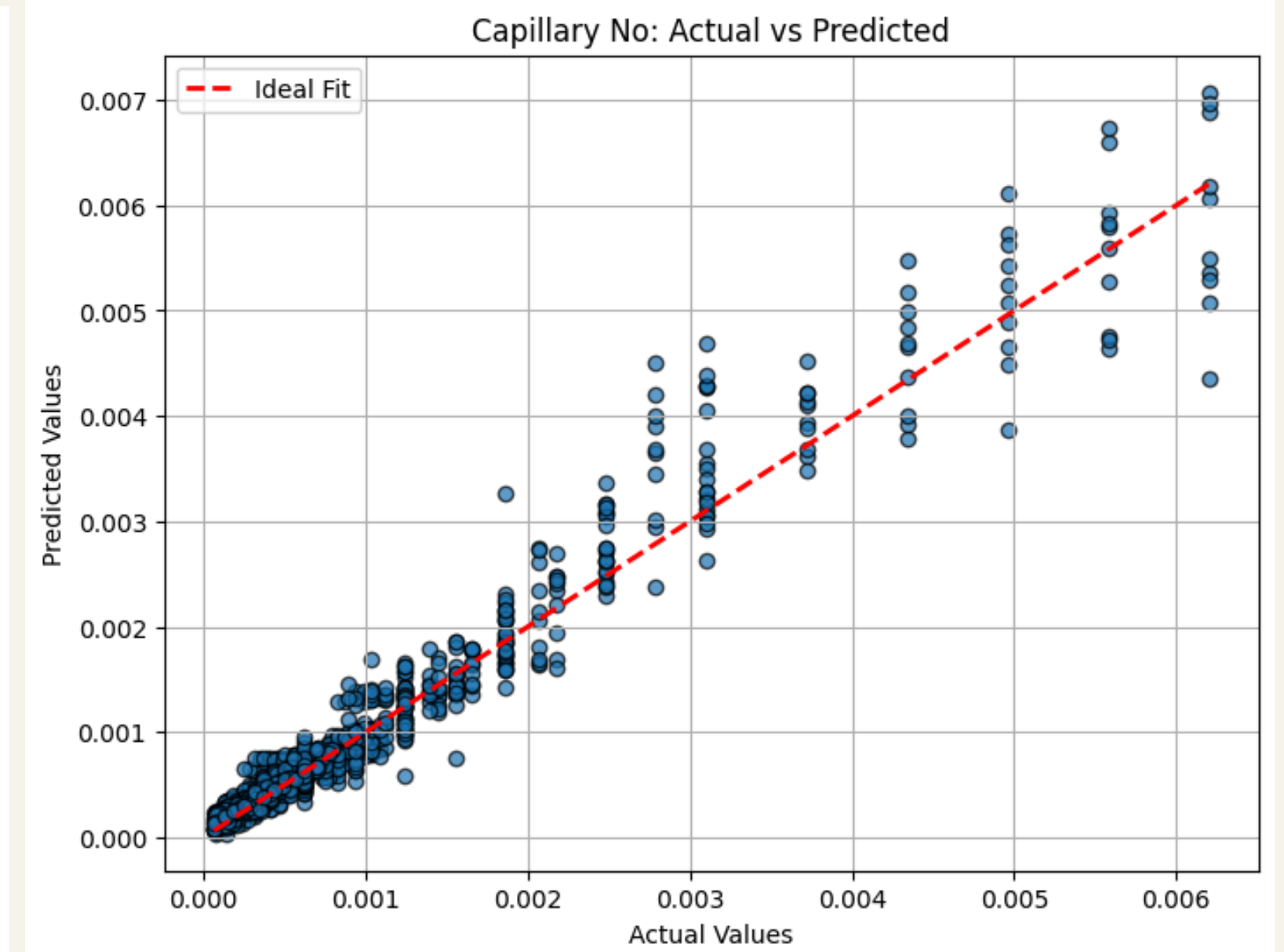
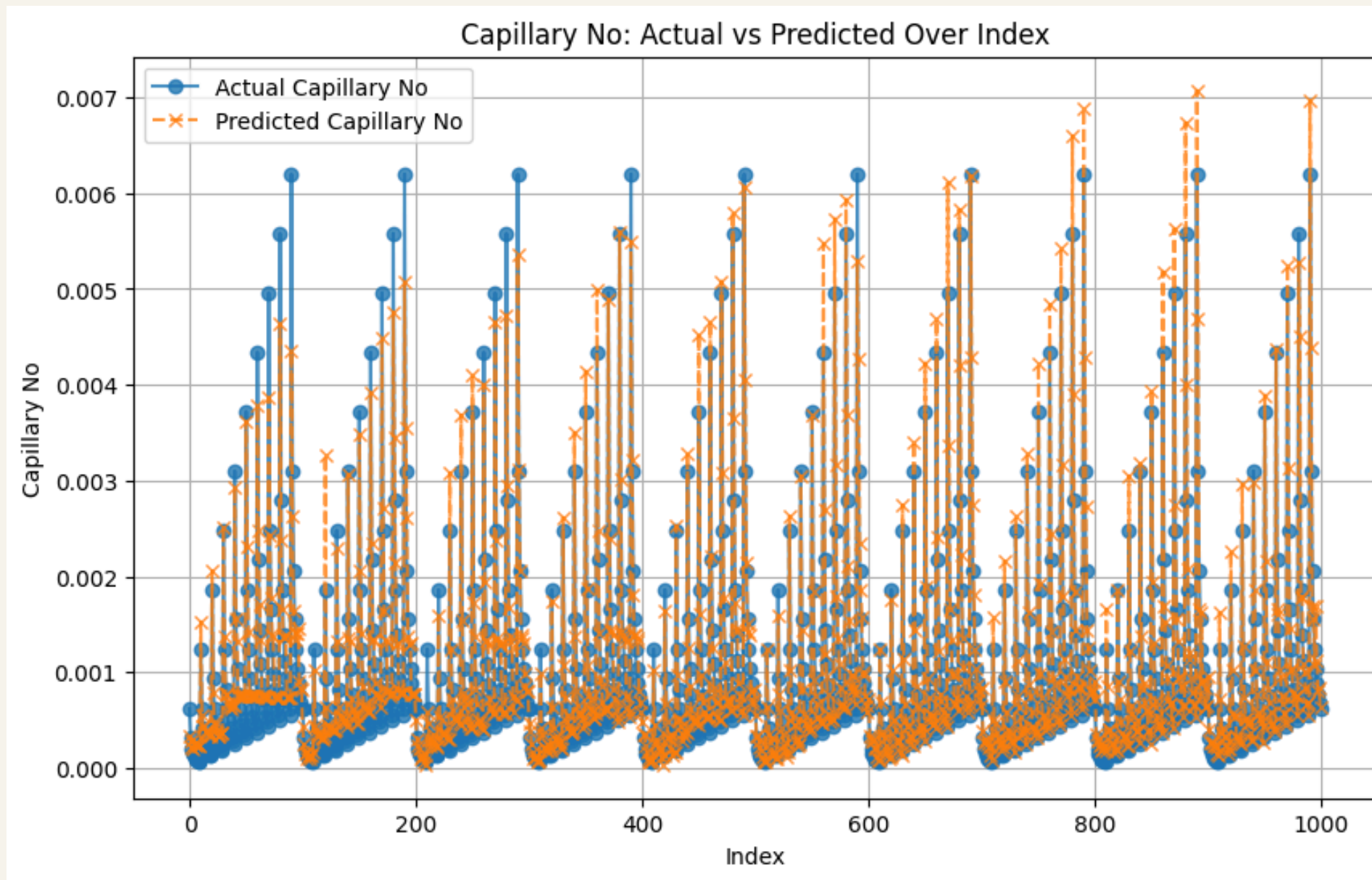
RESULTS - WEBER NUMBER

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RESULTS - CAPILLARY NUMBER

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The background features three vertical bars on the left: a wide pink one, a medium blue one, and a narrow beige one. The right side is a light beige field with two rectangular areas of small pink dots, one in the top right and one in the bottom right.

RECOMMENDATIONS

RECOMMENDATIONS

We have successfully estimated the flow rate ratio, weber number and capillary number of a system using image data. Now we suggest to build a heuristic for calculating droplet size using these parameters.

Based on the successful training and testing of our model for T-junction geometries, we suggest extending this study to explore Y-intersections and other complex geometrical configurations for further insights and broader applicability

We have done our analysis for two-phase flow. Now, we suggest extending it to more phases, particularly, a system containing three-phase flow, for further insights and a broader set of applications

REFERENCES

SkLearn Library Documentation: <https://scikit-learn.org/stable/>

TensorFlow Library Documentation: <https://www.tensorflow.org>

Keras Documentation: <https://faroit.com/keras-docs/1.2.0/>

NumPy Library Documentation: <https://numpy.org/doc/2.1/user/index.html>

Pandas Library Documentation: https://pandas.pydata.org/docs/user_guide/index.html

Matplotlib Documentation: <https://matplotlib.org/stable/index.html>

Seaborn Documentation: <https://seaborn.pydata.org/>

Wikipedia: <https://www.wikipedia.org/>

The background features three vertical stripes on the left: a wide pink stripe, a medium blue stripe, and a narrow beige stripe. The right side of the image is a light beige background with two rectangular areas of small, light pink dots in the top right and bottom right corners.

THANK YOU